WEKA Classification Assignment 01

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Objective and Dataset Exploration

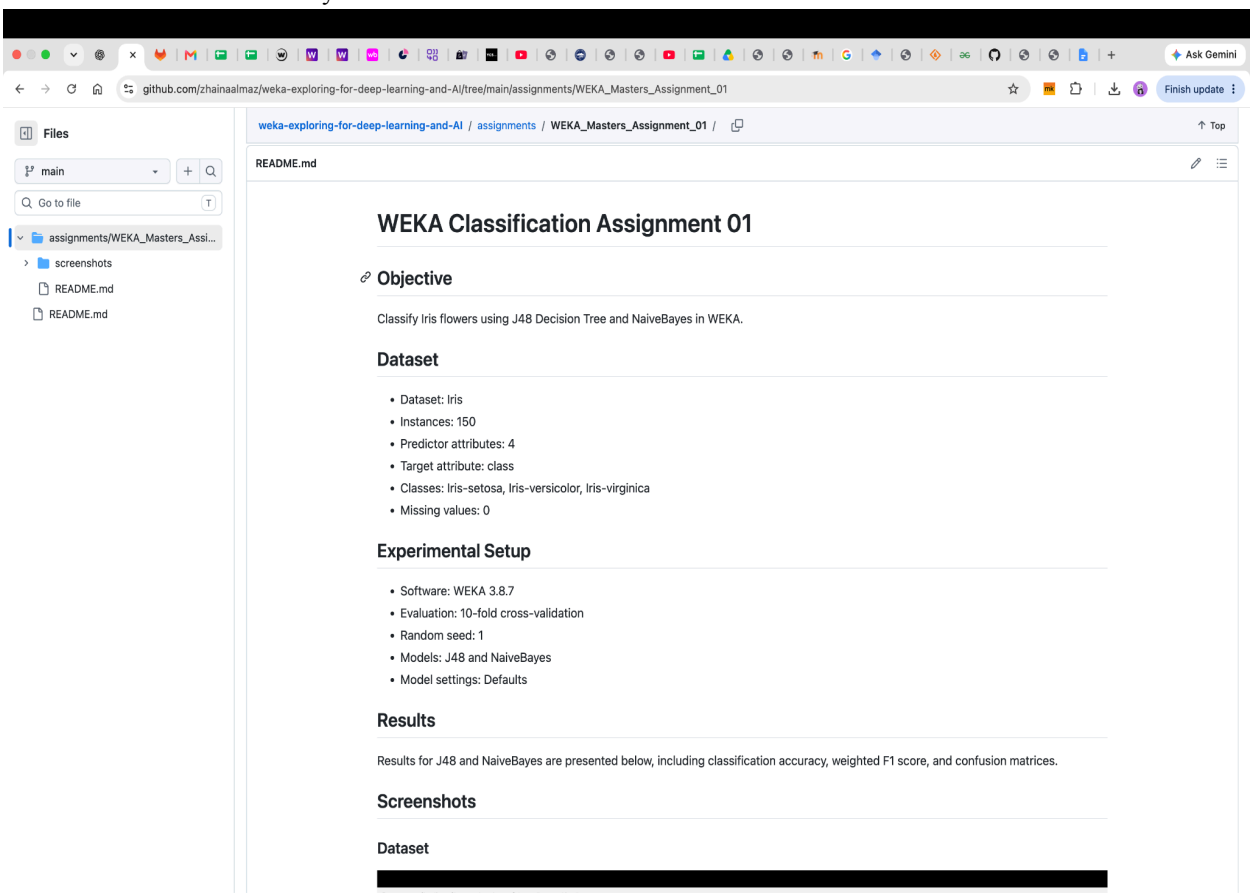
The objective was to classify Iris flowers using J48 Decision Tree and NaiveBayes in WEKA Explorer and compare the models using a fair evaluation.

Dataset: Iris (150 instances); 4 predictor attributes (sepal length, sepal width, petal length, petal width); target attribute: class; 3 classes: Iris-setosa, Iris-versicolor, Iris-virginica; 50 instances per class; 0 missing values.

Software and setup: WEKA 3.8.7; default settings for J48 (C=0.25, M=2) and NaiveBayes; 10-fold cross-validation; random seed 1; all attributes retained; no filters applied.

Scatter plot observation (petal length vs. petal width): Iris-setosa forms a distinct cluster with smaller petals and is clearly separated from the other classes. Iris-versicolor generally occupies the middle range, while Iris-virginica tends to have larger and wider petals. Some overlap between versicolor and virginica may contribute to their misclassification.

Screenshot 1. Dataset summary in WEKA

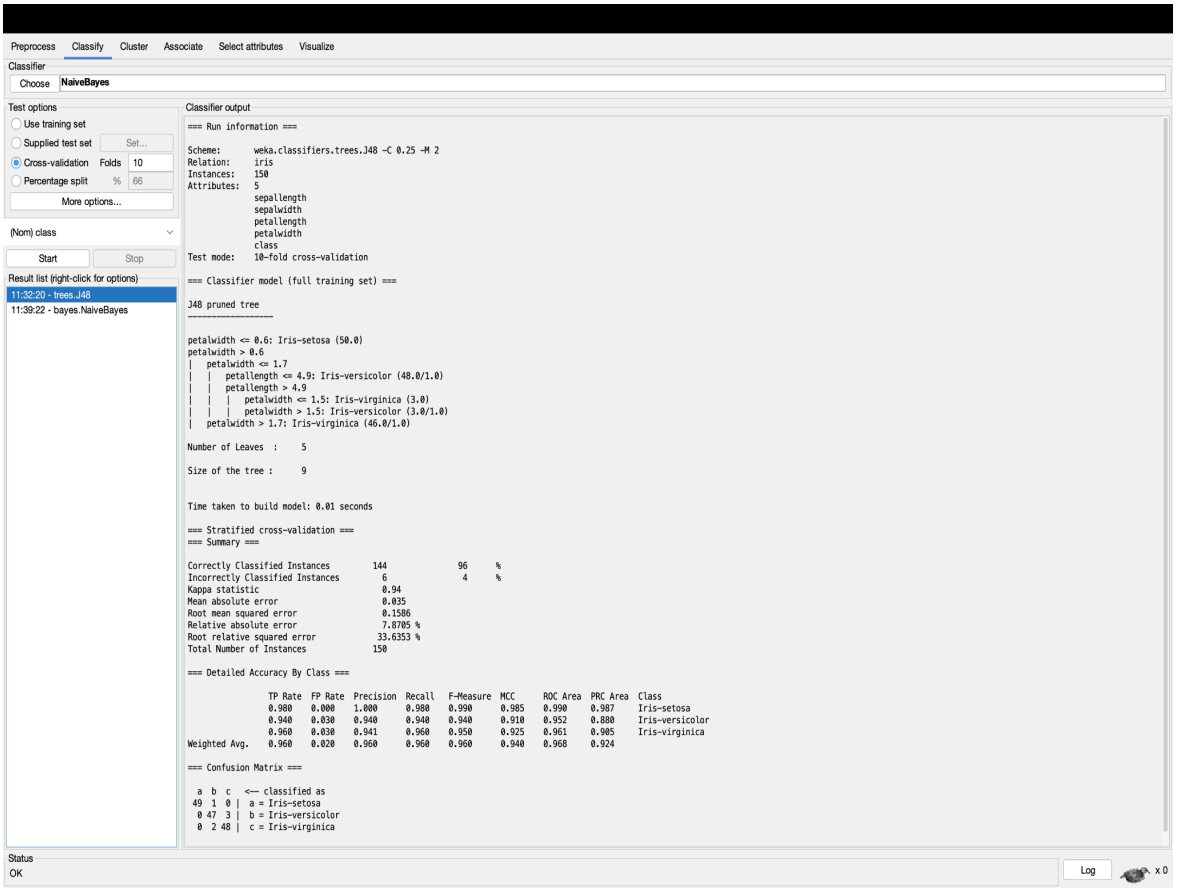


Model Evaluation Results

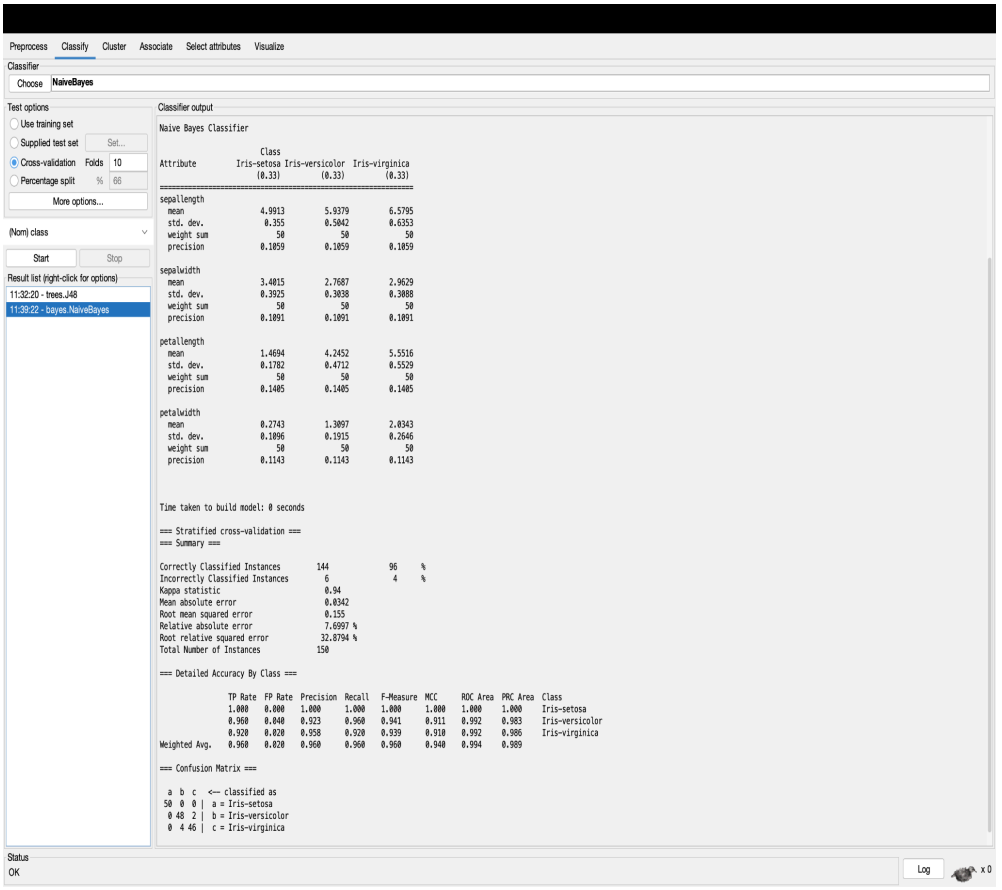
Both models were evaluated using 10-fold cross-validation with random seed 1.

Measure	J48	NaiveBayes
Accuracy (%)	96.00	96.00
Weighted average F1 score	0.960	0.960
Correctly classified	144/150	144/150
Incorrectly classified	6/150	6/150

Screenshot 2. J48 cross-validation evaluation output



Screenshot 3. NaiveBayes cross-validation evaluation output



Interpretation and Critical Comparison

Confusion matrix interpretation (J48)

J48 correctly classified 144 of 150 instances. The most confused pair was Iris-versicolor and Iris-virginica: three versicolor instances were classified as virginica, while two virginica instances were classified as versicolor. In addition, one Iris-setosa instance was classified as versicolor.

One rule from the displayed full-data J48 tree is: if petal width is less than or equal to 0.6, classify the flower as Iris-setosa. The displayed full-data tree is separate from the models fitted within cross-validation.

Critical comparison

J48 and NaiveBayes achieved the same performance on the Iris dataset: 96.00% accuracy and a weighted F1 score of 0.960. Based on these two measures, neither model outperformed the other. I recommend J48 when interpretability is important because its decision tree expresses predictions as explicit rules that can be inspected and explained. NaiveBayes is also efficient and provides probabilistic predictions, but its reasoning is less directly represented as a small set of decision rules. The confusion matrices show that both models mainly confuse Iris-versicolor and Iris-virginica, which have more overlapping petal measurements than Iris-setosa. Evaluating a model on the same data used to train it can overestimate performance because the model has already seen those examples and may fit patterns specific to them. Cross-validation provides a more reliable estimate by repeatedly evaluating models on held-out folds. However, this result comes from one small dataset and one cross-validation run. Different samples, datasets, or evaluation settings may produce different results, so these scores do not establish that J48 is universally better. The recommendation is therefore based on interpretability, not superior measured accuracy.

GitHub project: https://github.com/zhainaalmaz/weka-exploring-for-deep-learning-and-AI